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Module 1.1 The Economic Problem and Market Systems

1. The Economic Problem

1.1 Scarcity

Scarcity is the fundamental economic problem: human wants are unlimited, but resources (factors of production) are finite. This means choices must be made about how to allocate resources.

Scarcity applies to all economies — whether developed or developing. Even the wealthiest nations cannot produce everything their citizens desire. Scarcity forces three fundamental questions:

- **What** to produce? (Which goods and services should resources be allocated to?)
- **How** to produce? (What combination of resources and techniques should be used?)
- **For whom** to produce? (How should output be distributed among the population?)

1.2 Factors of Production

Resources used to produce goods and services are classified into four factors of production:

Factor	Definition	Reward	Examples
Land	All natural resources — both renewable and non-renewable	Rent	Oil, timber, water, farmland, minerals
Labour	The physical and mental effort of workers used in production	Wages	Doctors, factory workers, teachers

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Capital	Man-made aids to production (not money itself)	Interest	Machinery, factories, tools, computers
Enterprise	The factor that organises the other three and bears risk	Profit	Business owners, start-up founders

Exam tip: Capital in economics refers to physical/produced goods used in production, not financial capital (money). A factory is capital; the money used to buy it is not.

1.3 Choice and Opportunity Cost

Opportunity cost is the value of the next best alternative forgone when a choice is made. It applies to individuals, firms, and governments.

Because of scarcity, every decision has an opportunity cost. Examples:

- **Individual:** A student choosing to attend university forgoes 3 years of potential earnings (the opportunity cost).
- **Firm:** A company investing in new machinery forgoes the returns it could have earned investing elsewhere.
- **Government:** Spending more on defence means less available for healthcare or education.

Case Study: UK Government Spending Trade-offs (2020s)
During the COVID-19 pandemic, the UK government spent over £370 billion on pandemic support (furlough, business loans, NHS capacity). The opportunity cost was enormous — infrastructure projects were delayed, education budgets were squeezed, and public debt rose to over 100% of GDP, constraining future spending choices. The Crossrail project and HS2 both faced budget cuts and delays partly due to pandemic spending priorities.

1.4 Free Goods vs Economic Goods

- **Economic goods** — have an opportunity cost; scarce; e.g. cars, food, clothing

- **Free goods** — no opportunity cost; unlimited in supply relative to demand; e.g. sunlight, air (though clean air is increasingly an economic good)

2. Production Possibility Frontier (PPF)

Production Possibility Frontier (PPF) shows the maximum possible output combinations of two goods that an economy can produce when all resources are fully and efficiently employed, given current technology.

2.1 PPF Diagram

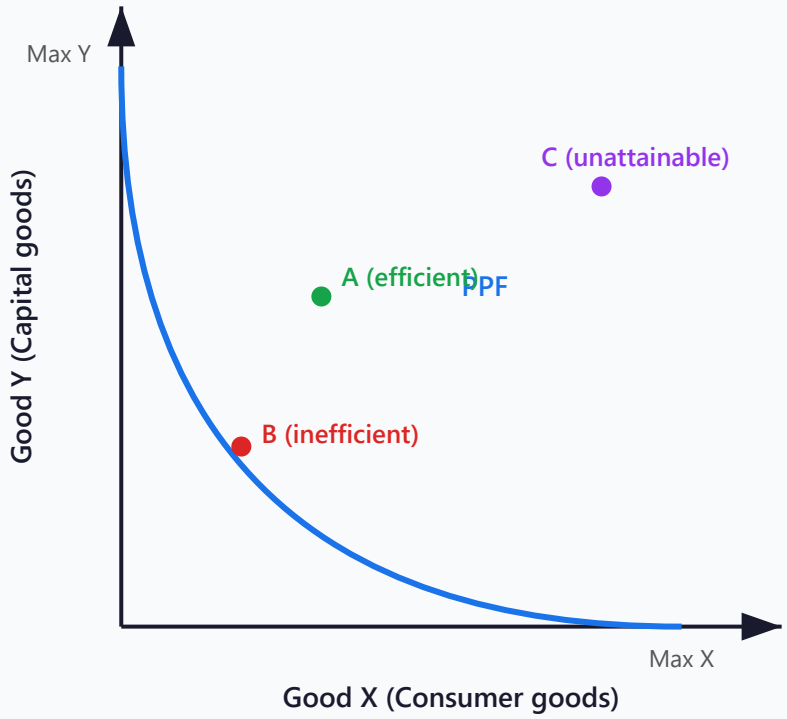

Figure 1: Production Possibility Frontier (quarter-circle shape showing increasing opportunity cost)

2.2 Key Points on the PPF

- **On the curve (Point A):** Productively efficient — all resources fully employed.

- **Inside the curve (Point B):** Inefficient — resources are unemployed or underutilised (e.g. during a recession).
- **Outside the curve (Point C):** Currently unattainable — would require more resources or better technology.

2.3 Movements Along vs Shifts of the PPF

Movement along the PPF represents a reallocation of resources from one good to another (illustrates opportunity cost).

Outward shift of the PPF represents economic growth — caused by:

- Increase in the quantity of factors of production (e.g. immigration increasing labour supply)
- Improvement in quality of factors (e.g. better education improving labour productivity)
- Technological advancement (e.g. automation increasing output per worker)

Inward shift represents a decline in productive capacity — caused by natural disasters, war, emigration, or resource depletion.

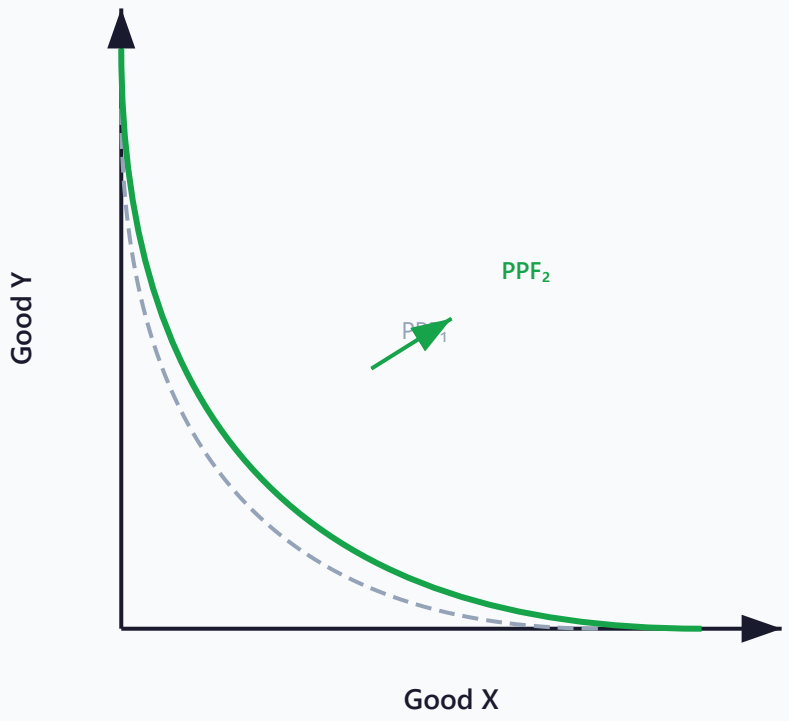

Figure 2: Outward shift of the PPF — representing economic growth

2.4 Opportunity Cost and the PPF

The concave (bowed-out) shape of the PPF illustrates **increasing opportunity cost**. As an economy produces more of one good, it must give up increasing amounts of the other because resources are not perfectly substitutable between uses.

A straight-line PPF would indicate **constant opportunity cost** — resources are equally productive in both uses (unrealistic in most cases).

Case Study: China's Economic Growth and PPF Shift

China's rapid industrialisation from the 1980s onwards represents a dramatic outward shift of the PPF. Massive investment in physical capital, a huge expansion of the educated workforce, and technology transfer from foreign firms all contributed. China's real GDP grew at an average of nearly 10% per year for three decades (1980-2010), shifting from a primarily agricultural economy to the world's largest manufacturer.

3. Economic Systems

3.1 Free Market Economy

In a **free market economy**, resources are allocated through the price mechanism with minimal government intervention.

- Private ownership of factors of production
- Prices determined by supply and demand
- Consumers are sovereign — their spending decisions signal to producers what to make
- Profit motive drives producers
- Competition between firms drives efficiency and innovation

Advantages: Efficient allocation through price signals, incentives for innovation, consumer choice, flexible response to changing conditions.

Disadvantages: Market failure (externalities, public goods under-provided), inequality of income/wealth, instability (boom-bust cycles), merit goods under-consumed.

3.2 Command (Planned) Economy

In a **command economy**, the government owns the factors of production and makes all allocation decisions centrally.

- State ownership of resources
- Central planning board decides what, how, and for whom to produce
- Prices set by the state rather than markets
- Aims to achieve equitable distribution

Advantages: Can reduce inequality, can provide public and merit goods, can avoid wasteful duplication, can plan for long-term objectives.

Disadvantages: Information problems (planners cannot know all consumer preferences), lack of incentives (no profit motive), inefficiency and surpluses/shortages, lack of consumer choice, bureaucracy.

3.3 Mixed Economy

A **mixed economy** combines elements of both free market and command economies. Most modern economies are mixed. The private sector operates through markets, while the government intervenes to correct market failures, provide public goods, and redistribute income.

The UK is a mixed economy: most goods are allocated through markets, but the government provides the NHS, education, defence, and welfare, funded through taxation.

Case Study: North Korea vs South Korea

North Korea operates as one of the few remaining command economies. The state controls virtually all production and distribution. South Korea, by contrast, operates a mixed market economy with strong private enterprise. The results are stark: South Korea's GDP per capita is estimated to be roughly 25 times that of North Korea. South Korea is a global technology leader (Samsung, Hyundai), while North Korea suffers from chronic food shortages. This comparison powerfully illustrates the impact of economic systems on living standards.

4. The Price Mechanism

The price mechanism is the means by which decisions of consumers and producers interact to determine the allocation of resources in a market economy through the forces of supply and demand.

4.1 Three Functions of the Price Mechanism

Function	Explanation	Example
Signalling	Prices convey information to buyers and sellers. Rising prices signal growing scarcity or rising demand; falling prices signal surplus or falling demand.	Rising house prices signal high demand and limited supply in a region.
Incentive	Price changes create incentives for economic agents to change behaviour. High prices incentivise producers to supply more and consumers to buy less.	High oil prices incentivise investment in renewable energy.
Rationing	Prices ration scarce resources. When supply is limited, prices rise, excluding those unable or unwilling to pay, allocating goods to those who value them most (in monetary terms).	Concert tickets for popular artists rise in price, rationing seats to those willing to pay most.

4.2 Adam Smith and the Invisible Hand

Adam Smith argued that individuals pursuing their own self-interest are led, as if by an "invisible hand," to promote the good of society. Producers seeking profit produce what consumers want. Consumers seeking value buy at the lowest prices. Competition ensures efficient allocation without central planning.

5. Demand

Demand is the quantity of a good or service that consumers are willing and able to purchase at a

given price in a given time period. Effective demand requires both willingness and ability to pay.

5.1 The Law of Demand

There is an **inverse (negative) relationship** between price and quantity demanded, ceteris paribus (all other things being equal). As price rises, quantity demanded falls, and vice versa.

This is because of:

- **Income effect:** When price rises, consumers' real income falls (they can afford less), so they buy less.
- **Substitution effect:** When price rises, consumers switch to cheaper alternatives.
- **Diminishing marginal utility:** Each additional unit consumed provides less extra satisfaction, so consumers will only buy more at lower prices.

5.2 Demand Curve

Figure 3: A demand curve — downward sloping showing the inverse relationship between price and quantity demanded

5.3 Movements Along vs Shifts of the Demand Curve

A **movement along** the demand curve is caused only by a change in the good's own price. This is called a change in **quantity demanded**.

A **shift** of the demand curve is caused by a change in a non-price factor. This is called a change in **demand**.

Factors that Shift Demand (Non-Price Determinants)

- **Income:** For normal goods, a rise in income shifts demand right. For inferior goods, demand shifts left.
- **Price of related goods:** Substitutes — if the price of a substitute rises, demand for this good rises. Complements — if the price of a complement rises, demand for this good falls.
- **Tastes and preferences:** Advertising, fashion trends, health scares can shift demand.
- **Population size and structure:** A growing population increases demand; an ageing population shifts demand toward healthcare.
- **Expectations:** Expected future price rises can increase current demand (e.g. buying before a tax increase).
- **Interest rates:** Lower rates increase demand for goods bought on credit (houses, cars).

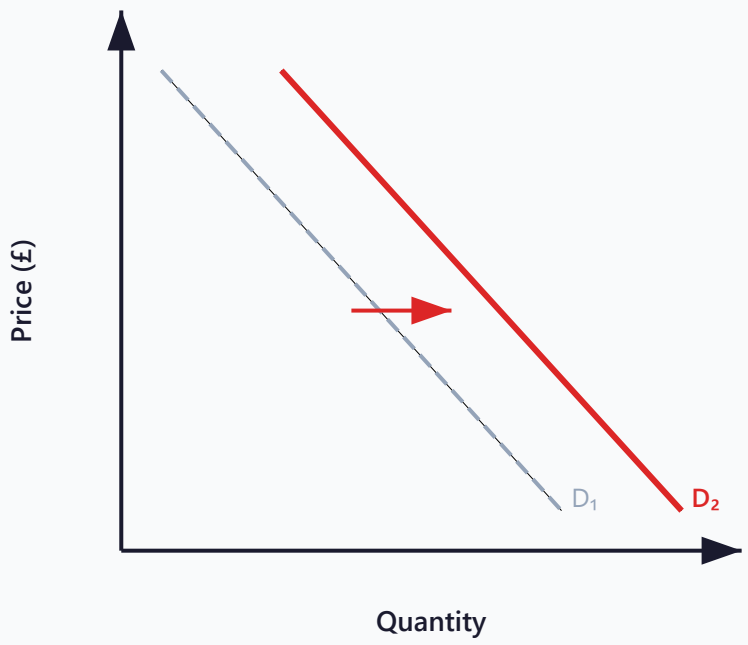

Figure 4: Rightward shift in demand (increase in demand)

6. Supply

Supply is the quantity of a good or service that producers are willing and able to offer for sale at a given price in a given time period.

6.1 The Law of Supply

There is a **positive (direct) relationship** between price and quantity supplied, ceteris paribus. As price rises, quantity supplied rises because:

- Higher prices make production more profitable, incentivising existing firms to produce more
- Higher prices attract new firms to enter the market
- Higher prices allow firms to cover the rising marginal costs of extra output

6.2 Supply Curve

Figure 5: A supply curve — upward sloping showing the positive relationship between price and quantity supplied

6.3 Factors that Shift Supply (Non-Price Determinants)

- **Costs of production:** Rising wages, raw material costs, or energy prices shift supply left (decrease). Falling costs shift supply right.
- **Technology:** Improved technology reduces unit costs and shifts supply right.
- **Taxes and subsidies:** An indirect tax shifts supply left (raises costs). A subsidy shifts supply right (lowers costs).
- **Number of firms:** More firms entering the market increases market supply (rightward shift).
- **Weather/natural factors:** Particularly important for agricultural markets. Good harvests shift supply right.
- **Productivity:** Improved worker productivity reduces unit costs and shifts supply right.

7. Market Equilibrium

Equilibrium is the point where the quantity demanded equals the quantity supplied at a given price. There is no tendency to change — the market clears with no excess supply or demand.

7.1 Equilibrium Diagram

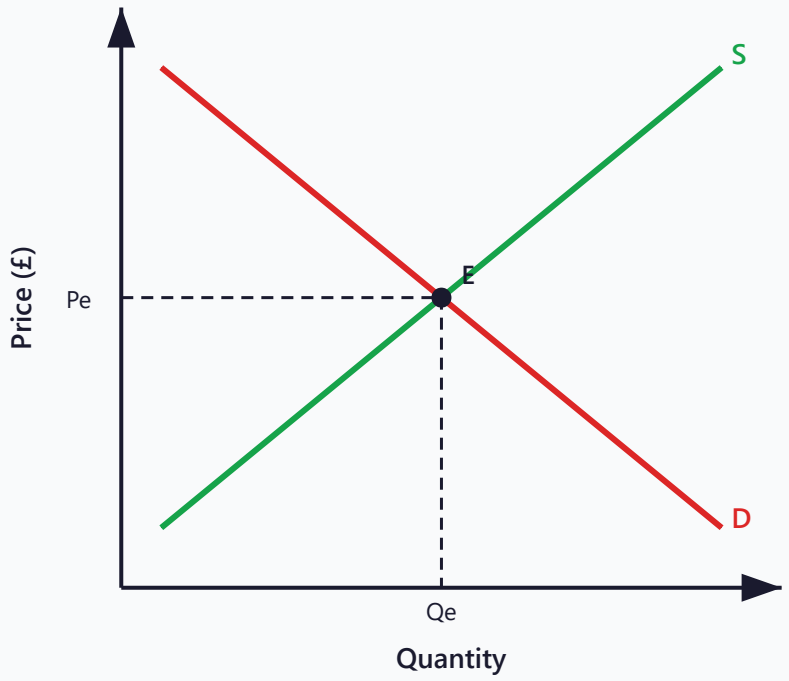

Figure 6: Market equilibrium where demand equals supply at Pe, Qe

7.2 Excess Supply and Excess Demand

- **Excess supply (surplus):** When price is above equilibrium, Qs > Qd. Unsold stock accumulates, putting downward pressure on price until equilibrium is restored.
- **Excess demand (shortage):** When price is below equilibrium, Qd > Qs. Consumers compete for limited goods, putting upward pressure on price until equilibrium is restored.

7.3 Shifts in Demand and Supply — Effects on Equilibrium

Change	Effect on Equilibrium Price	Effect on Equilibrium Quantity
Increase in demand (D shifts right)	Rises	Rises
Decrease in demand (D shifts left)	Falls	Falls
Increase in supply (S shifts right)	Falls	Rises
Decrease in supply (S shifts left)	Rises	Falls

Exam tip: When both demand and supply shift simultaneously, one of either the price or quantity effect will be indeterminate (depends on the relative size of the shifts). Always state this in your answer.

8. Consumer and Producer Surplus

Consumer surplus is the difference between what consumers are willing to pay and what they actually pay. It is the area below the demand curve and above the market price.

Producer surplus is the difference between the price producers receive and the minimum price they would be willing to accept. It is the area above the supply curve and below the market price.

8.1 Surplus Diagram

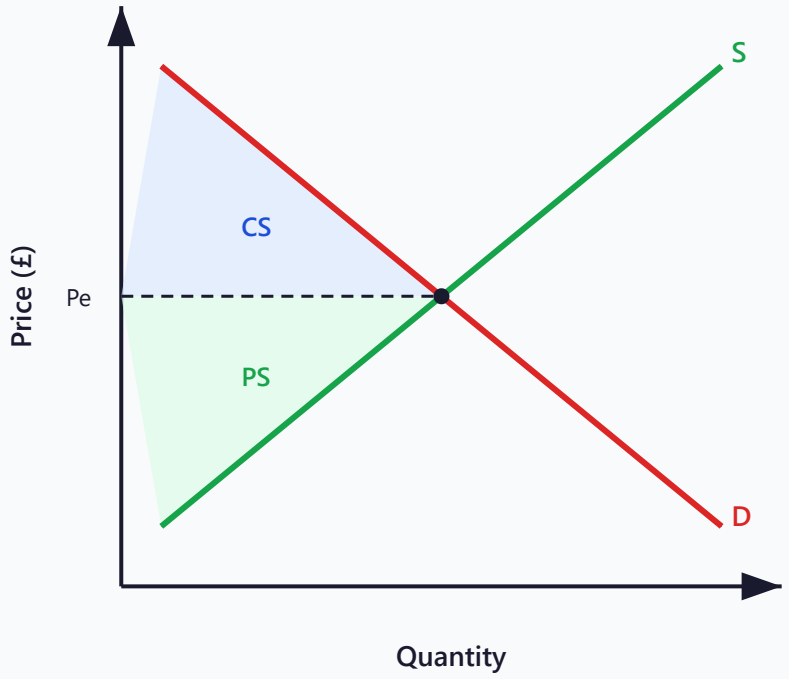

Figure 7: Consumer surplus (CS) and producer surplus (PS) at market equilibrium

8.2 Changes in Surplus

- An **increase in supply** (S shifts right) lowers price, increasing consumer surplus and possibly decreasing producer surplus.
- An **increase in demand** (D shifts right) raises price, increasing producer surplus and possibly decreasing consumer surplus.
- Total economic welfare = consumer surplus + producer surplus. This is maximised at the free market equilibrium (assuming no market failure).

9. Elasticity

9.1 Price Elasticity of Demand (PED)

Price Elasticity of Demand (PED) measures the responsiveness of quantity demanded to a change in the good's own price.

$$\text{PED} = \%\text{ change in quantity demanded} / \%\text{ change in price}$$

PED is always **negative** (due to the inverse relationship), but we typically refer to the absolute value.

Value	Description	Meaning
|PED| > 1	Elastic	Quantity demanded changes proportionally more than price. A 10% price rise leads to a >10% fall in Qd.
|PED| < 1	Inelastic	Quantity demanded changes proportionally less than price. A 10% price rise leads to a <10% fall in Qd.
|PED| = 1	Unit elastic	Percentage change in Qd exactly equals percentage change in price.
PED = 0	Perfectly inelastic	Vertical demand curve — quantity demanded does not respond to price changes.
PED = ∞	Perfectly elastic	Horizontal demand curve — any price rise above the given level causes demand to fall to zero.

Determinants of PED

- **Availability of substitutes:** More substitutes = more elastic (consumers can switch easily).
- **Necessity vs luxury:** Necessities (insulin, bread) tend to be inelastic; luxuries (holidays) elastic.
- **Proportion of income:** Goods that take a large share of income are more elastic.
- **Time period:** Demand is more elastic in the long run (more time to find alternatives).
- **Habit/addiction:** Addictive goods (cigarettes) tend to be inelastic.
- **Brand loyalty:** Strong branding makes demand less elastic.

PED and Total Revenue

- If demand is **elastic** (|PED| > 1): a price cut increases total revenue (the gain from extra sales outweighs the revenue lost per unit).
- If demand is **inelastic** (|PED| < 1): a price rise increases total revenue (the loss of sales is proportionally small).
- Total revenue is maximised when |PED| = 1.

Exam tip: The relationship between PED and total revenue is a very commonly tested concept. Always link elasticity to revenue when discussing pricing decisions by firms or the impact of indirect taxes.

9.2 Price Elasticity of Supply (PES)

Price Elasticity of Supply (PES) measures the responsiveness of quantity supplied to a change in price.

$$\text{PES} = \%\text{ change in quantity supplied} / \%\text{ change in price}$$

PES is always **positive** (supply curves slope upward).

Determinants of PES

- **Spare capacity:** Firms with spare capacity can increase output quickly (more elastic).
- **Availability of stocks:** Firms holding large inventories can respond quickly to price changes.
- **Time period:** Supply is more elastic in the long run (firms can build new factories, enter/exit market).
- **Factor mobility:** If resources can be easily switched between uses, supply is more elastic.
- **Nature of the good:** Agricultural goods have inelastic supply in the short run (growing seasons). Manufactured goods can be produced more quickly.

9.3 Income Elasticity of Demand (YED)

Income Elasticity of Demand (YED) measures the responsiveness of demand to a change in consumer income.

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$$\text{YED} = \%\text{ change in quantity demanded} / \%\text{ change in income}$$

Value	Type of Good	Explanation
YED > 0	Normal good	Demand rises as income rises
YED > 1	Luxury good (superior)	Demand rises more than proportionally as income rises (e.g. designer clothing, fine dining)
0 < YED < 1	Necessity	Demand rises less than proportionally as income rises (e.g. bread, milk, toothpaste)
YED < 0	Inferior good	Demand falls as income rises (e.g. supermarket value ranges, bus travel)

Case Study: Luxury Goods and Income Elasticity — LVMH

LVMH (Louis Vuitton Moet Hennessy) revenues have grown strongly alongside rising global incomes, particularly among the expanding middle classes in China and India. Between 2019 and 2023, LVMH revenue rose from approximately €53bn to €86bn, reflecting the high income elasticity of demand for luxury goods (YED well above 1). Conversely, during recessions, luxury firms typically see disproportionately large falls in demand.

9.4 Cross Elasticity of Demand (XED)

Cross Elasticity of Demand (XED) measures the responsiveness of demand for good A to a change in the price of good B.

$$\text{XED} = \%\text{ change in quantity demanded of Good A} / \%\text{ change in price of Good B}$$

Value	Relationship	Example

XED > 0 (positive)	Substitutes	Coca-Cola and Pepsi — if Pepsi's price rises, demand for Coca-Cola rises
XED < 0 (negative)	Complements	Printers and ink cartridges — if printer prices rise, demand for ink falls
XED = 0	Unrelated goods	Bread and televisions — price of one has no effect on demand for the other

The **magnitude** of XED indicates the strength of the relationship. A high positive XED means the goods are close substitutes.

9.5 Significance of Elasticity for Firms and Government

- **Pricing strategies:** Firms with inelastic demand can raise prices to increase revenue. Firms with elastic demand may lower prices to boost sales.
- **Tax incidence:** The burden of an indirect tax falls more heavily on the side of the market that is more inelastic. If demand is inelastic, consumers bear most of the tax burden.
- **Government revenue:** Taxes on goods with inelastic demand (cigarettes, petrol) raise more revenue because consumption falls little.
- **Agricultural policy:** Agricultural supply is inelastic in the short run, making prices volatile. Government may intervene with buffer stocks or subsidies.

Case Study: UK Sugar Tax (Soft Drinks Industry Levy, 2018)

The UK introduced a tiered sugar tax on soft drinks in April 2018. Before introduction, many manufacturers reformulated their drinks to reduce sugar content (shifting to below the tax thresholds). Drinks like Ribena and Lucozade reduced sugar by 50%. The tax raised approximately £340 million in its first year but less than initially forecast, partly because reformulation was so widespread. The relatively elastic supply response (firms reformulating) was arguably the policy's greatest success — it changed producer behaviour even before the tax took full effect. Consumer demand for full-sugar drinks proved somewhat elastic, with shifts toward sugar-free alternatives.

Evaluation — The Price Mechanism

- The price mechanism allocates resources efficiently in many markets, but fails where there are externalities, public goods, imperfect information, or monopoly power.
- Elasticity analysis helps explain why some markets respond well to price signals while others (e.g. addictive goods, necessities) do not.
- Government intervention (taxes, subsidies, regulation) may improve outcomes but can also cause government failure.
- Real-world markets are rarely perfectly competitive — market power, asymmetric information, and behavioural biases mean the invisible hand does not always work perfectly.

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